

CÉCILE SOUDÉ

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Biomedical Engineer · AI for biology

EDUCATION

MSc Biomedical Engineering, Imperial College London

London, UK | Oct 2025 – Aug 2026

Relevant coursework: Data Analysis and Statistics, Machine Learning, Computational Fluid Mechanics, Molecular, Cell and Tissue Engineering, Human Neuromechanical Learning, Medical Devices Certifications, Physiology

MEng General Engineering, Ecole Centrale de Lyon

Lyon, France | Sept 2022 – Aug 2026

Relevant coursework: Mathematics, Computer Science, Image Processing, Physics, Health Engineering, Mechanical Engineering, Materials Engineering, Computational Fluid Dynamics, Economics

Advanced Mathematics and Physics Program, Lycée Charlemagne

Paris, France | Sept 2019 – July 2022

Preparation for national competitive entrance exams to leading French 'Grandes Ecoles'

Intensive program in advanced mathematics and physics

EXPERIENCES

Research Engineer, Computer Vision & Single-Molecule Science, EMBL x UNSW Sydney | Feb – Aug 2025

- Built from scratch a computer vision pipeline (Python, OpenCV) for Single-Particle Tracking using fluorescence microscopy. Used advanced image processing techniques (Gaussian fitting, noise filtering, subpixel refinement) to overcome low signal-to-noise ratio images, achieving nanometer-scale localization accuracy
- Upgraded existing diagnostic equipment for Parkinson's disease research by designing an integrated control system and UI, directly improving user biologists experience and experimental precision

R&D Biomechanic Test Laboratory Intern, Stryker

Grenoble, France | May – Dec 2024

Project Manager – Research test campaign on Stryker's reverse shoulder prosthesis

- Applied structured and creative thinking to plan and define each stage of the research project, assessing how bone substitute materials affect Stryker implant performance, with results guiding future testing methods
- Led testing validations within a COFRAC-accredited laboratory, ensuring strict compliance with FDA and Europe MDR regulatory standards

COMPUTER SCIENCE & DATA PROJECTS

Master Thesis, Research in Human Performance, Imperial College London London, UK | Jan – Aug 2026

- Analysed multi-modal time-series data (high-frequency EMG, 3D Motion Capture, and force plates) to identify biomechanical patterns in high-variability gait cycles using Python and MATLAB scripts

AI & Healthcare Hackathon Finalist, Mistral AI x GE Healthcare

Lyon, France | Feb 2026

- Built a multi-agent AI workflow integrating RAG pipelines and long/short-term memory to ground LLMs on oncology medical guidelines (RECIST 1.1) to produce hallucination-free reports for tumor MRI scans
- Designed a predictive clinical triage strategy using XGBoost and Learning-to-Rank models to replace traditional FIFO hospital queues, optimizing patient flow
- Pitched the solution to a jury of healthcare executives and AI experts, ranking in the top 8 out of 23 teams

LEADERSHIP & VOLUNTEERING EXPERIENCES

Vice-President of Centrale Lyon's Sports Office

Lyon, France | Jan 2023 – Jan 2024

- Managed an 11-student team to organize sports tournaments and drive sports engagement on campus

Member of the Event Team in CheerUp! Centrale Lyon

Lyon, France | Oct 2023 – Oct 2024

- Visited young cancer patients and organized a charity race (€1.5k raised) to fund and fulfil their dreams

TECHNICAL SKILLS & LANGUAGES

Programming: Python (Advanced: OpenCV, NumPy, scikit-learn, pandas), github, MATLAB (Advanced)

Languages: French (native), English (C1, TOEFL iBT 104/120), Japanese (rudiments)

Sports: Judo black belt with 18 years of practice at a highly competitive level, international judo camp training and assistant coach for 4 to 5 years old children / Football: Striker in Centrale Lyon Women's team

Music: 8 years of guitar in a conservatory: obtained the second cycle of classical guitar with honours